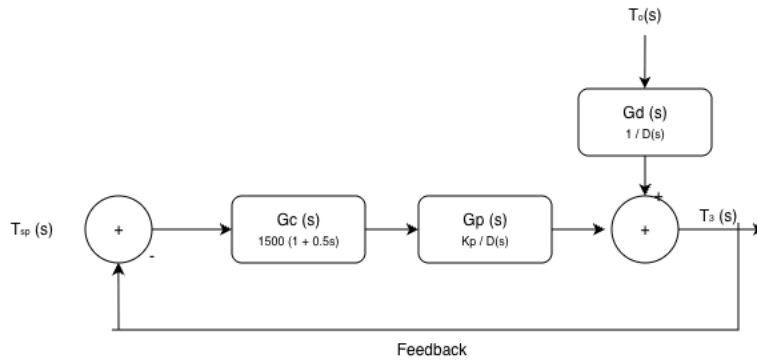


Problem 1

Part 1:



Part 2:

The open loop transfer function is the product of the process and Proportional-Derivative controller

The denominator polynomial common to all three tanks:

$$D(s) = (s + 1)(1.25s + 1)(1.5s + 1) = 1.875s^3 + 4.625s^2 + 3.75s + 1$$

The process gain:

$$K_p = \frac{1}{\omega C_p} = \frac{1}{(250)(1)} = 0.004 \frac{^\circ \cdot \text{min}}{\text{Btu}}$$

The open loop transfer function is:

$$\begin{aligned} G_c(s) \cdot G_p(s) &= 1500(1 + 0.5s) \cdot \frac{0.004}{1.875s^3 + 4.625s^2 + 3.75s + 1} \\ &= \frac{3s + 6}{1.875s^3 + 4.625s^2 + 3.75s + 1} \end{aligned}$$

Applying the closed loop formula then adding $3s + 6$ to the denominator:

$$\frac{T_3(s)}{T_{sp}(s)} = \frac{3s + 6}{1.875s^3 + 4.625s^2 + 6.75s + 7}$$

Part 3:

The inlet temperature enters through the same three tanks making its transfer function to T3:

$$G_d(s) = \frac{1}{1.875s^3 + 4.625s^2 + 3.75s + 1}$$

With a fixed set point, the closed-loop disturbance transfer function comes to:

$$\frac{T_3(s)}{T_o(s)} = \frac{G_d}{1 + G_c G_p} = \frac{\frac{1}{D(s)}}{\frac{1.875s^3 + 4.625s^2 + 6.75s + 7}{D(s)}}$$

This then comes to:

$$\frac{T_3(s)}{T_o(s)} = \frac{1}{1.875s^3 + 4.625s^2 + 6.75s + 7}$$

Part 4:

For a unit-step disturbance $T_o(s) = \frac{1}{s}$ with the set point unchanged. The T3 can be found using the Final Value Theorem:

$$T_3(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{1}{1.875s^3 + 4.625s^2 + 6.75s + 7} \cdot \frac{1}{s} = \frac{1}{7} \approx 0.1429^\circ F$$

Since the set point deviation is zero, the offset is:

$$\text{Offset} = T_{sp} - T_3(\infty) = 0 - \frac{1}{7}$$

$$\text{Offset} = -\frac{1}{7} \approx -0.1429^\circ F$$

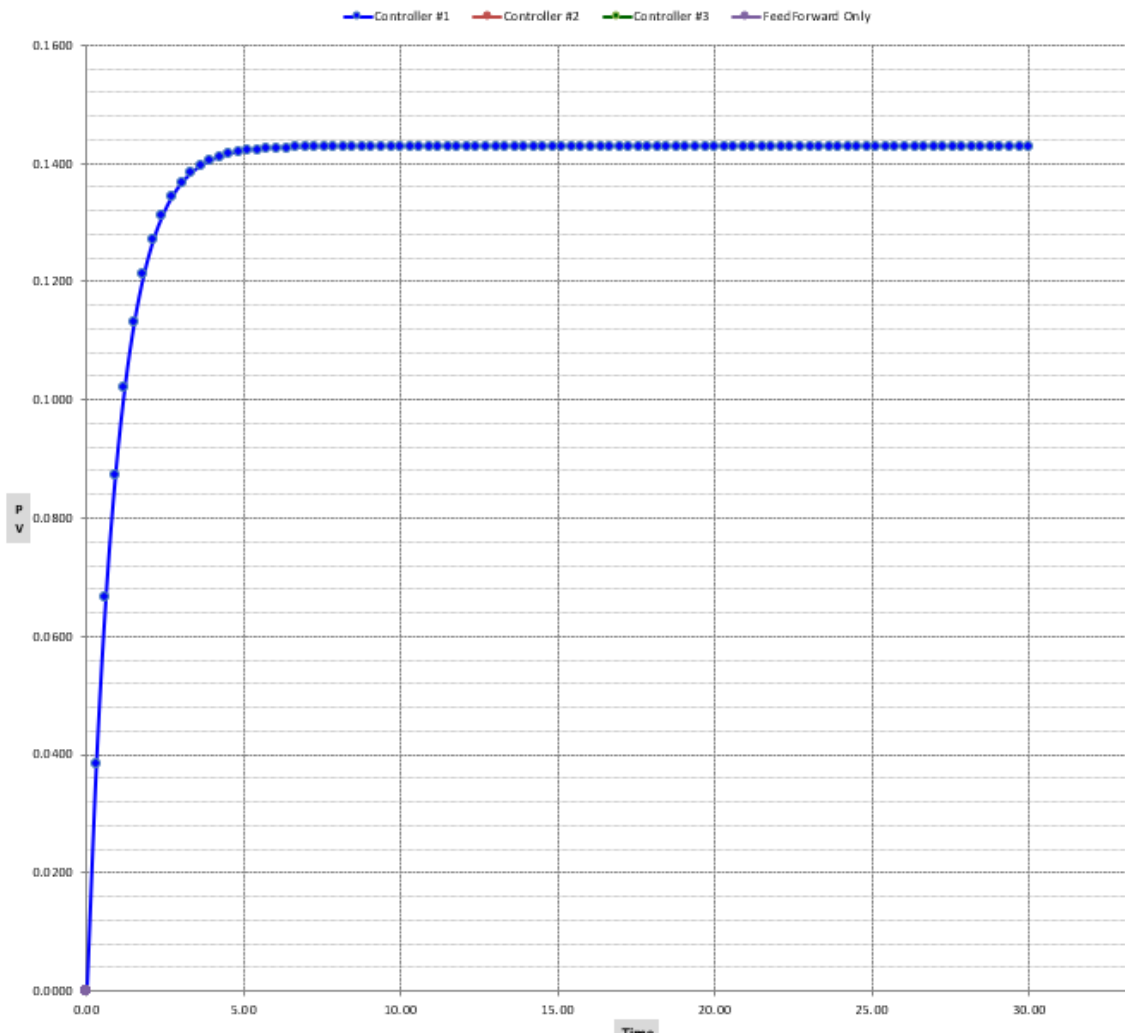
The negative sign shows that the T3 settles above the set point by 0.1429 F. This offset is expected; a Proportional Derivative controller includes no integral action and can't move the steady-state error to zero while a sustained disturbance occurs.

Part 5:

The closed-loop system with the Proportional-derivative control was simulated using the CHE Calculator for a unit step change in the inlet temperature. The controller and process parameters were entered to get the resulting graph.

Process gain (Kp) was 0.004 °F·min/Btu, the process lag time was 3.75 minutes, the process time delay was 0 minutes, the disturbance gain was 1, the disturbance lag time was 3.75 minutes, the controller gain (Kc) was 1500 Btu/min/°F and the derivative time is 0.5 minutes. There is no integral time, and the disturbance change is 1.

Process Performance:PV



Controller Retuning		CONTROLLER #1			CONTROLLER #2			CONTROLLER #3			Feedforward CONTROLLER			CASCADE CONTROLLER			SETTLING TIMES			
Plant Data	Model Plant	Time	PV	CO	Time	PV	CO	Time	PV	CO	Time	PV	CO	Time	PV	CO	Cr	Cr	Cr	Cr
		0.0000	0.0000	-101.111							0.0000	0.0000	0.0000				0.0000			0.0000
		0.3030	0.0000	-108.312							0.0000	0.0000	0.0000				0.0000			0.0000
		0.6060	0.0000	-105.256							0.0000	0.0000	0.0000				0.0000			0.0000
		0.9090	0.0000	-102.093							0.0000	0.0000	0.0000				0.0000			0.0000
		1.2121	0.0000	-104.312							0.0000	0.0000	0.0000				0.0000			0.0000
		1.5151	0.0000	-102.047							0.0000	0.0000	0.0000				0.0000			0.0000
		1.8181	0.0000	-108.639							0.0000	0.0000	0.0000				0.0000			0.0000
		2.1212	0.0000	-102.093							0.0000	0.0000	0.0000				0.0000			0.0000
		2.4242	0.0000	-105.345							0.0000	0.0000	0.0000				0.0000			0.0000
		2.7272	0.0000	-108.185							0.0000	0.0000	0.0000				0.0000			0.0000
		3.0303	0.0000	-109.312							0.0000	0.0000	0.0000				0.0000			0.0000
		3.3333	0.0000	-110.322							0.0000	0.0000	0.0000				0.0000			0.0000
		3.6363	0.0000	-111.509							0.0000	0.0000	0.0000				0.0000			0.0000
		3.9394	0.0000	-112.550							0.0000	0.0000	0.0000				0.0000			0.0000
		4.2424	0.0000	-113.083							0.0000	0.0000	0.0000				0.0000			0.0000
		4.5454	0.0000	-113.260							0.0000	0.0000	0.0000				0.0000			0.0000
		4.8484	0.0000	-113.096							0.0000	0.0000	0.0000				0.0000			0.0000
		5.1515	0.0000	-113.719							0.0000	0.0000	0.0000				0.0000			0.0000
		5.4545	0.0000	-113.321							0.0000	0.0000	0.0000				0.0000			0.0000
		5.7575	0.0000	-114.023							0.0000	0.0000	0.0000				0.0000			0.0000
		6.0606	0.0000	-114.093							0.0000	0.0000	0.0000				0.0000			0.0000
		6.3636	0.0000	-114.142							0.0000	0.0000	0.0000				0.0000			0.0000
		6.6667	0.0000	-114.301							0.0000	0.0000	0.0000				0.0000			0.0000
		6.9697	0.0000	-114.230							0.0000	0.0000	0.0000				0.0000			0.0000
		7.2727	0.0000	-114.230							0.0000	0.0000	0.0000				0.0000			0.0000
		7.5757	0.0000	-114.245							0.0000	0.0000	0.0000				0.0000			0.0000
		7.8787	0.0000	-114.256							0.0000	0.0000	0.0000				0.0000			0.0000
		8.1818	0.0000	-114.264							0.0000	0.0000	0.0000				0.0000			0.0000
		8.4848	0.0000	-114.270							0.0000	0.0000	0.0000				0.0000			0.0000
		8.7879	0.0000	-114.274							0.0000	0.0000	0.0000				0.0000			0.0000
		9.0909	0.0000	-114.275							0.0000	0.0000	0.0000				0.0000			0.0000
		9.3939	0.0000	-114.276							0.0000	0.0000	0.0000				0.0000			0.0000
		9.6970	0.0000	-114.282							0.0000	0.0000	0.0000				0.0000			0.0000
		10.0000	0.0000	-114.282							0.0000	0.0000	0.0000				0.0000			0.0000
		10.3030	0.0000	-114.283							0.0000	0.0000	0.0000				0.0000			0.0000
		10.6061	0.0000	-114.284							0.0000	0.0000	0.0000				0.0000			0.0000
		10.9091	0.0000	-114.284							0.0000	0.0000	0.0000				0.0000			0.0000
		11.2121	0.0000	-114.284							0.0000	0.0000	0.0000				0.0000			0.0000
		11.5151	0.0000	-114.284							0.0000	0.0000	0.0000				0.0000			0.0000
		11.8181	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		12.1212	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		12.4242	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		12.7272	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		13.0303	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		13.3333	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		13.6364	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		13.9394	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		14.2424	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000
		14.5454	0.0000	-114.285							0.0000	0.0000	0.0000				0.0000			0.0000

The simulation confirmed the results from Part 4. The temperature T3 rose then settled around 0.1429 °F above the set point which is line to an offset of -1/7 °F. This offset is consistent with the lack of integral action in a PD controller as the controller can reduce error but not eliminate the error under a sustained disturbance.

Part 6

To remove the offset, a PID controller was designed using the Ziegler-Nichols tuning method. To determine the ultimate and period, the Routh-Hurwitz stability criterion was used.

$$1.875s^3 + 4.625s^2 + 3.75s + (1 + 0.004K_{cu}) = 0$$

$$-1.875\omega u^3 + 3.75\omega u = 0 \rightarrow \omega u^2 = 2 \rightarrow \omega u = \sqrt{2} \text{ rad/min}$$

Ultimate Gain:

$$-4.625(2) + 1 + 0.004K_{cu} = 0$$

$$K_{cu} = 8.25 / 0.004 = 2062.5 \text{ Btu/min/°F}$$

Ultimate period:

$$T_u = 2\pi / \sqrt{2} = 4.443 \text{ min}$$

Ziegler-Nichols PID tuning:

$$K_c = 0.6 \times 2062.5 = 1237.5 \text{ Btu/min/°F}$$

$$\tau_I = 0.5 \times 4.443 = 2.22 \text{ min}$$

$$\tau_D = 0.125 \times 4.443 = 0.555 \text{ min}$$

PID TRANSFER FUNCTION

$$G_c(s) = 1237.5 \times (1 + 1/(2.22s) + 0.555s)$$

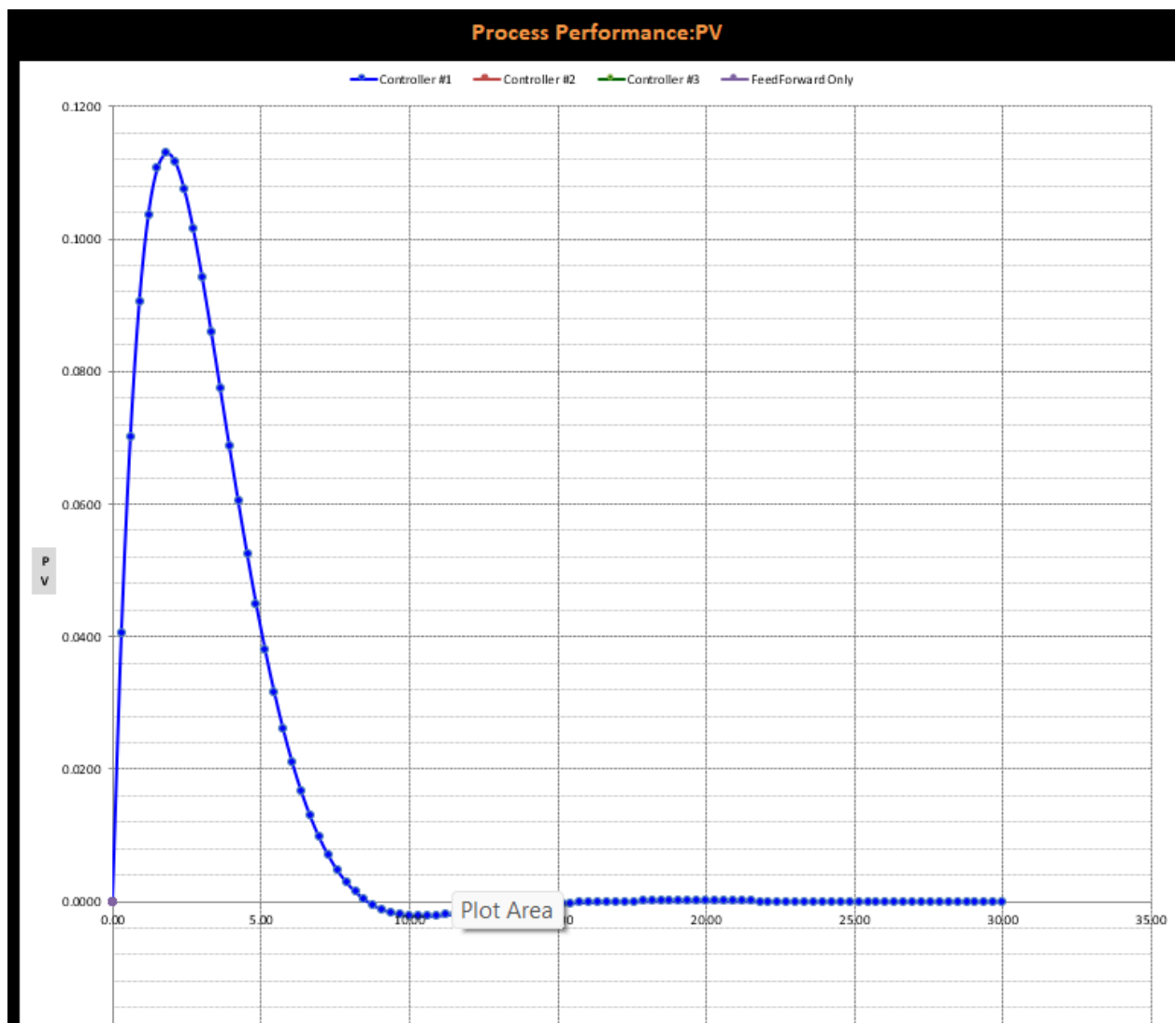
Final Value Theorem shows zero offset:

The PID controller contains integral action so the controller gain $G_c(s) \rightarrow \infty$ as $s \rightarrow 0$.

Applying the final value theorem to the closed-loop disturbance transfer function:

$$T_3(\infty) = \lim(s \rightarrow 0) [G_d / (1 + G_c \cdot G_p)] = 1 / (1 + \infty) = 0 \text{ } ^\circ\text{F}$$

The integrator moves the denominator to infinity at steady-state, making the offset zero.



Controller Retuning		CONTROLLER #1			CONTROLLER #2			CONTROLLER #3			FeedForward CONTROLLER			CASCADE CONTROLLER			SETTLING TIMES			
Read Data	Reset Point	Time	PV	CD	Time	PV	CD	Time	PV	CD	Time	PV	CD	Time	PV	CD	Controller #1	Controller #2	Controller #3	FF Controller
		0.0000	0.0000	-15.7082							0.0000	0.0000	0.0000				0.0000			0.0000
		0.3000	0.0405	-152.4245							0.0000	0.0000	0.0000				0.0405			0.0000
		0.6000	0.0701	-165.9107							0.0000	0.0000	0.0000				0.0701			0.0000
		0.9000	0.0906	-176.2804							0.0000	0.0000	0.0000				0.0906			0.0000
		1.2000	0.1036	-183.7214							0.0000	0.0000	0.0000				0.1036			0.0000
		1.5000	0.1107	-208.9082							0.0000	0.0000	0.0000				0.1107			0.0000
		1.8000	0.1100	-225.0385							0.0000	0.0000	0.0000				0.1100			0.0000
		2.1000	0.1185	-238.8171							0.0000	0.0000	0.0000				0.1185			0.0000
		2.4000	0.1075	-238.8739							0.0000	0.0000	0.0000				0.1075			0.0000
		2.7000	0.1004	-245.7027							0.0000	0.0000	0.0000				0.1004			0.0000
		3.0000	0.0941	-250.7567							0.0000	0.0000	0.0000				0.0941			0.0000
		3.3000	0.0899	-254.9652							0.0000	0.0000	0.0000				0.0899			0.0000
		3.6000	0.0774	-257.3395							0.0000	0.0000	0.0000				0.0774			0.0000
		3.9000	0.0688	-259.2768							0.0000	0.0000	0.0000				0.0688			0.0000
		4.2000	0.0604	-261.3951							0.0000	0.0000	0.0000				0.0604			0.0000
		4.5000	0.0523	-261.0023							0.0000	0.0000	0.0000				0.0523			0.0000
		4.8000	0.0448	-261.8222							0.0000	0.0000	0.0000				0.0448			0.0000
		5.1000	0.0379	-260.9761							0.0000	0.0000	0.0000				0.0379			0.0000
		5.4000	0.0316	-260.5339							0.0000	0.0000	0.0000				0.0316			0.0000
		5.7000	0.0260	-259.9098							0.0000	0.0000	0.0000				0.0260			0.0000
		6.0000	0.0210	-259.1448							0.0000	0.0000	0.0000				0.0210			0.0000
		6.3000	0.0167	-258.3489							0.0000	0.0000	0.0000				0.0167			0.0000
		6.6000	0.0129	-257.5016							0.0000	0.0000	0.0000				0.0129			0.0000
		6.9000	0.0097	-256.6536							0.0000	0.0000	0.0000				0.0097			0.0000
		7.2000	0.0070	-255.8286							0.0000	0.0000	0.0000				0.0070			0.0000
		7.5000	0.0048	-255.0435							0.0000	0.0000	0.0000				0.0048			0.0000
		7.8000	0.0030	-254.3102							0.0000	0.0000	0.0000				0.0030			0.0000
		8.1000	0.0015	-253.6363							0.0000	0.0000	0.0000				0.0015			0.0000
		8.4000	0.0004	-253.0259							0.0000	0.0000	0.0000				0.0004			0.0000
		8.7000	-0.0005	-252.4797							0.0000	0.0000	0.0000				0.0005			0.0000
		9.0000	-0.0015	-251.9986							0.0000	0.0000	0.0000				0.0015			0.0000
		9.3000	-0.0016	-251.5794							0.0000	0.0000	0.0000				0.0016			0.0000
		9.6000	-0.0019	-251.2164							0.0000	0.0000	0.0000				0.0019			0.0000
		9.9000	-0.0021	-250.9111							0.0000	0.0000	0.0000				0.0021			0.0000
		10.2000	-0.0022	-250.6532							0.0000	0.0000	0.0000				0.0022			0.0000
		10.5000	-0.0022	-250.4399							0.0000	0.0000	0.0000				0.0022			0.0000
		10.8000	-0.0021	-250.2654							0.0000	0.0000	0.0000				0.0021			0.0000
		11.1000	-0.0020	-250.1275							0.0000	0.0000	0.0000				0.0020			0.0000
		11.4000	-0.0019	-250.0288							0.0000	0.0000	0.0000				0.0019			0.0000
		11.7000	-0.0017	-249.9583							0.0000	0.0000	0.0000				0.0017			0.0000
		12.0000	-0.0016	-249.9150							0.0000	0.0000	0.0000				0.0016			0.0000

Part 7

The settling time from the simulation results showing for the PID controller is around 12 minutes

Parameter	PD Controller	PID Controller
Controller Gain, Kc (Btu/min/°F)	1500	1237.5
Integral Time (min)	-	2.22
Derivative Time (min)	0.5	0.555
Steady state offset (°F)	-0.143	0
Settling time (min)	N/A	~12
ITAE	64.15	1.40

The simulation confirms the theoretical predictions from part 4 and 6. The PD controller shows a permanent offset of -0.143 °F, indicating that T3 settles above the setpoint and remains there. Moving to a PID controller removes this. Initially T3 overshoots to 0.11 °F then returns to 0°F.

The ITAE values outline the performance difference. The PID controller has a 1.40 score and 64.15 score for the PD controller. The lower the ITAE score the better, the PID controller is better in terms of error accumulation over time. The main reason for this is integral action.

Problem B

Part A

To reject the feed disturbance, the feedforward must cancel the disturbance effect before it reaches the tray so setting the temperature change to zero gives:

$$GFF(s) = -GD / GP = - [20e^{(-15s)} / (30s+1)] / [-5e^{(-2s)} / (6s+1)]$$

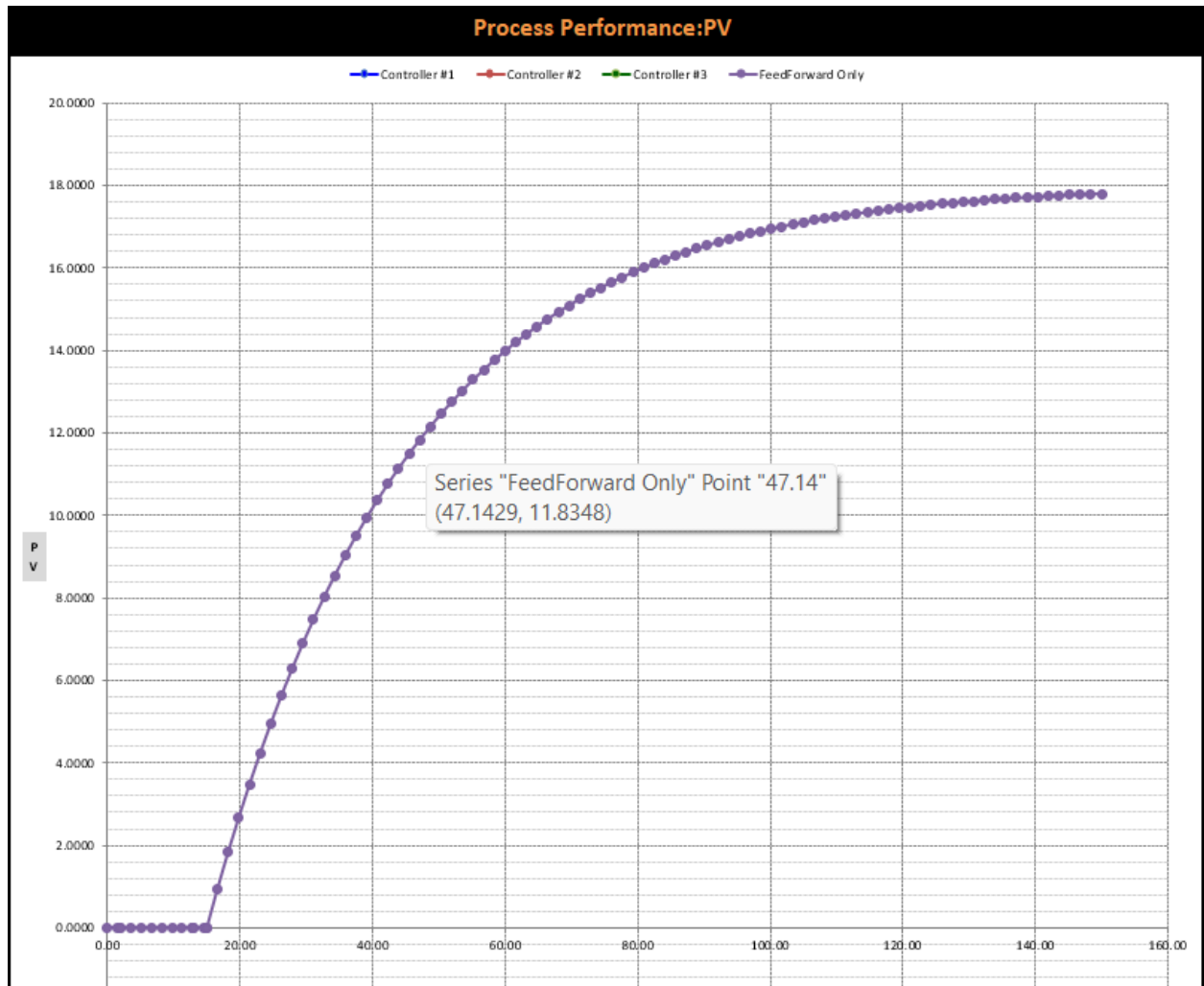
Coming to:

$$GFF(s) = 4 \times (6s+1) / (30s+1) \times e^{(-13s)}$$

Feedforward gain is 4, the feedforward dead time is 13 minutes, the lead time is 6 minutes, and the lag time taken from the disturbance lag is 30 minutes. It is **realizable** because the dead time (13 minutes) is positive.

Part B

The feedforward controller was applied to the process with a step change of 2 in the feed rate. The temperature stayed at zero until 15 minutes which is when the disturbance dead time passed, rose and then settled at 17.8 °F. This offset occurs because of the model mismatch. The disturbance gain (25) and process gain (4) are different from their assumed values which were used to design the controller, so a cancellation is not possible.



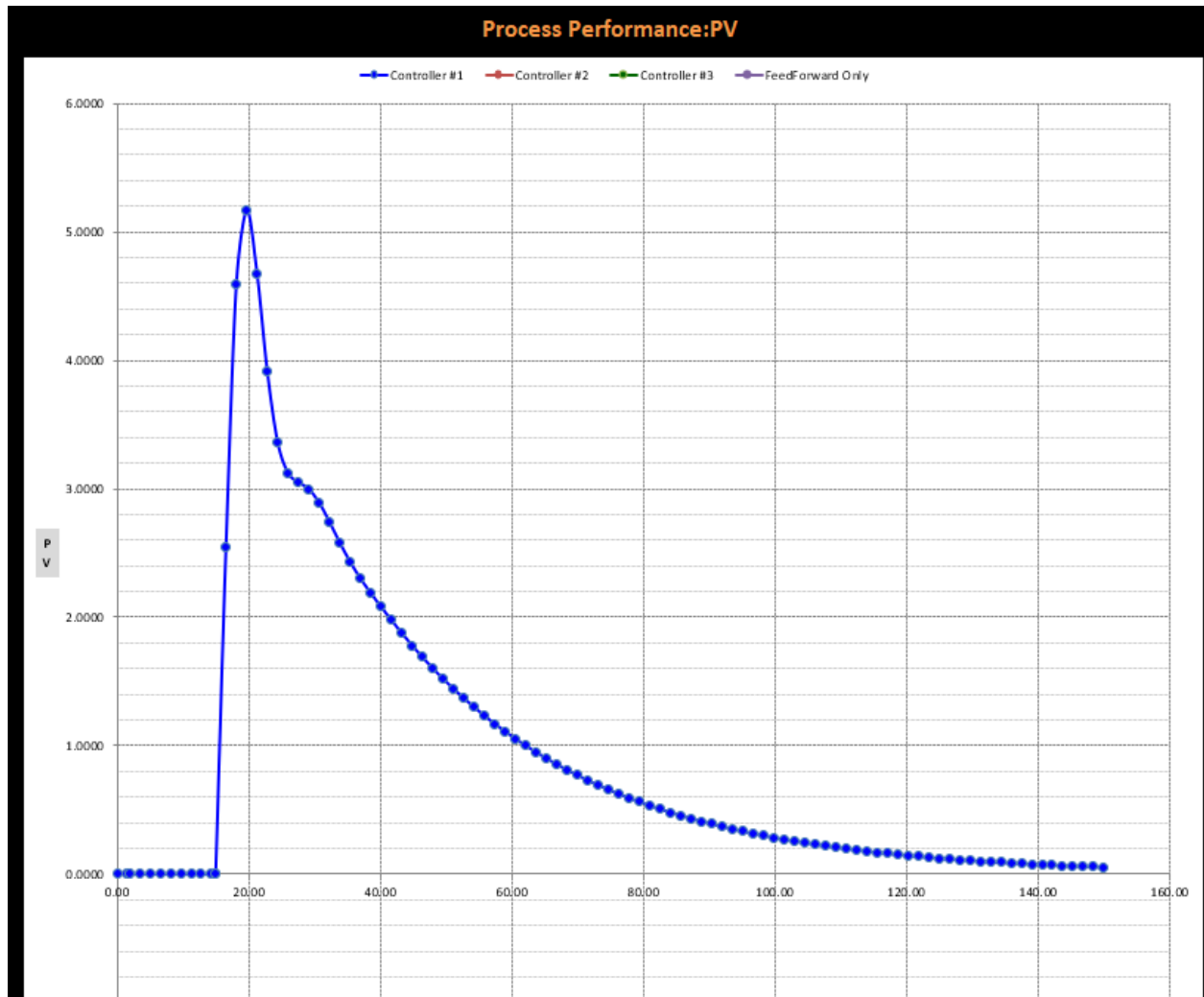
Part C

Here, a PI controller was designed using the model below:

$$K_c = \tau / [K_p(\lambda + \theta)] = 6 / [(-5)(0.25 + 2)] = -0.533 \text{ Btu/min/}^\circ\text{F}$$

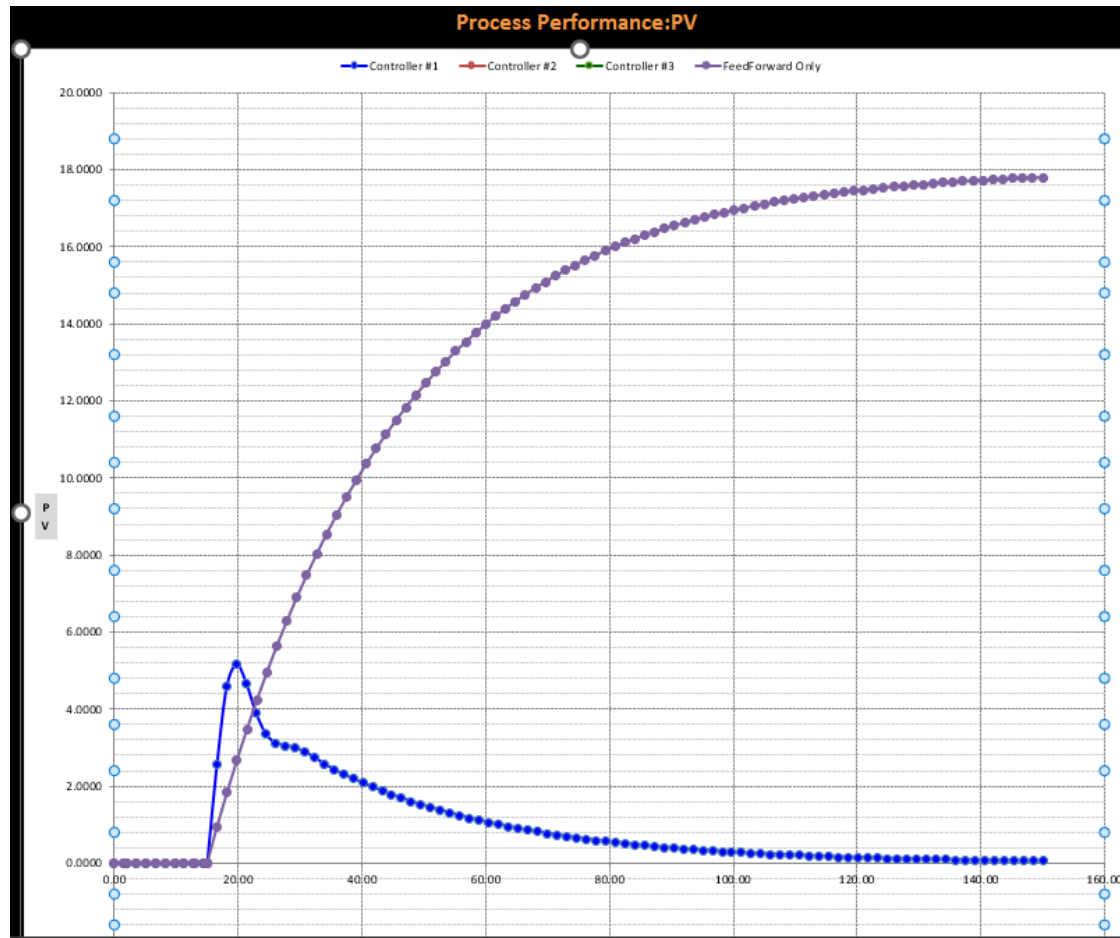
With the integral time being 6 minutes.

When applied to the process, the temperature sharply peaked at 5.3 °F around 18 minutes before the integral action brought it back to zero. Full settling took around 120 to 150 minutes and eventually the feedback controller removes the offset, but it only responded after the error came.



Part D

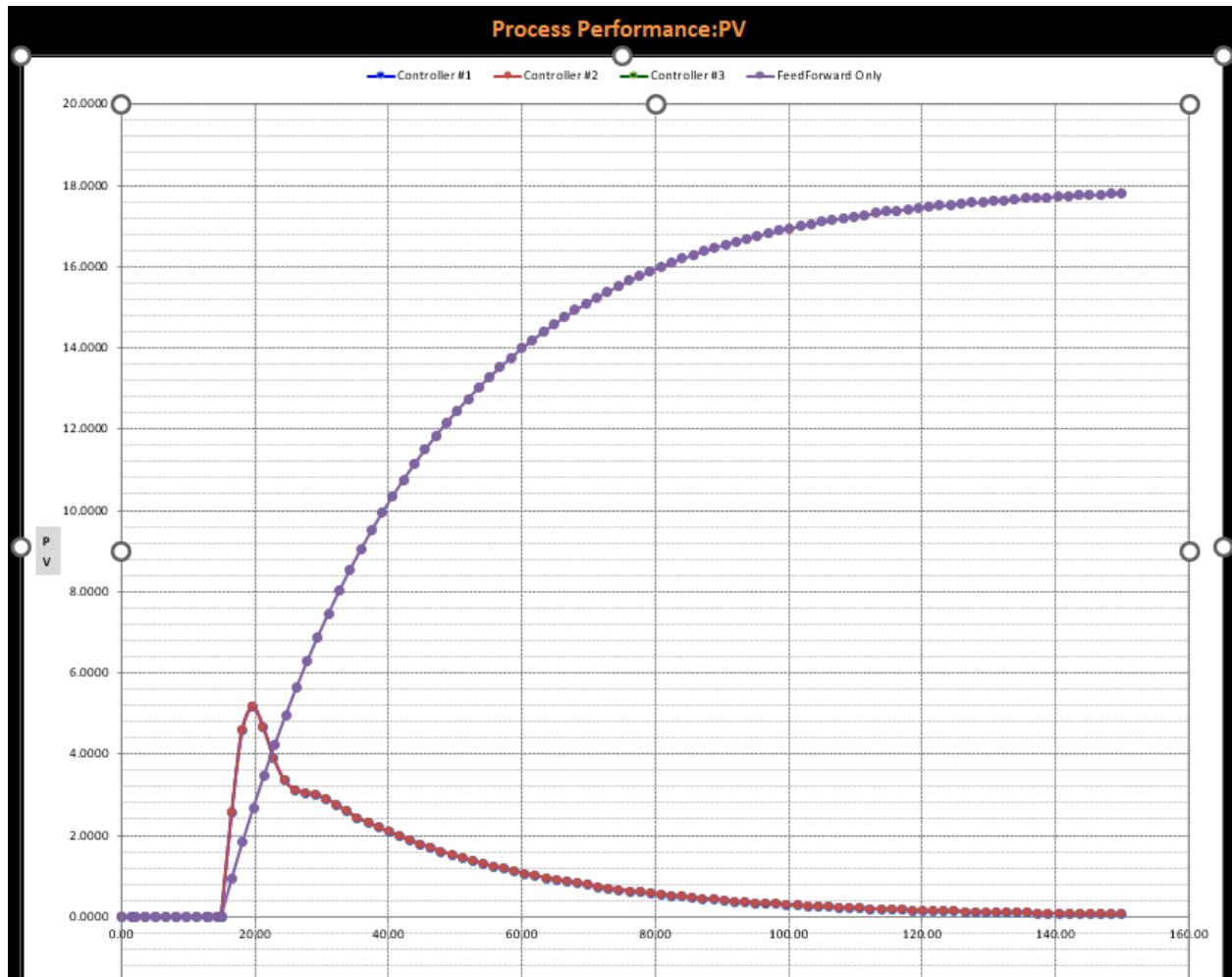
Both controllers were applied together to the process. The temperature peaked around 5 °F due to a model mismatch which reduced the feedforward controller's contribution, but then it returned to zero cleaner than when with the feedback controller alone. The feedback controller corrected whatever the feedforward controller left behind, and the integral action ensured no steady state offset.



Part E

	Deviation	Final Value	Observation
Feedforward only	17.8 °F	17.8°F	Quick in action but leaves big residual
Feedback only	5.3 °F	0°F	Removes offset but slow response
Feedforward with feedback	5.0 °F	0°F	Best choice overall, feedforward reduces early deviation and feedback cleans up the remaining residuals.

The feedforward controller didn't perform as expected. However, under a perfect model, it could have achieved zero offset, but the mismatch between actual gains and assumed meant a large residual was left behind. The combination of the feedforward and feedback scheme had the best result.



Note: Controller 1 (FF + FB are combined) and Controller 2 (FB alone) have the same feedback parameters which show their response overlapping on the graph. You can see the difference in the early transient region where the feedforward contribution from Controller 1 gives a faster initial correction.

References

Smith, C. A., & Corripio, A. B. (2005). *Principles and Practice of Automatic Process Control* (3rd ed.). John Wiley & Sons.

Caracotsios, M. (2024). *CHE Calculator Version 13: A Computational Tool for Chemical Process Control* [Excel-based software]. Department of Chemical Engineering, University of Illinois Chicago. Contact: mcaracot@uic.edu